

Maharashtra Public Service Commission Assessment

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Lecturer Electronics Maharashtra Education Service Group B Higher and Technical Education Department

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Lecturer Electronics Maharashtra Education Service Group B Higher and Technical Education Department

Section Id :	4410093087658
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Sub-Section Number :	1
Sub-Section Id :	4410093098683
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 4410092092447 Question Type : MCQ Option Shuffling : No

Which of the following components is NOT a part of a standard bridge-based temperature detector system?

Options :

1. A precision instrumentation amplifier stage
2. A thermistor or resistive temperature sensing element
3. A balanced Wheatstone bridge network
4. A high-pass filter with a 100 megahertz cutoff frequency

Question Number : 2 Question Id : 4410092092455 Question Type : MCQ Option Shuffling : No

For a practical non-inverting amplifier circuit utilizing an operational amplifier, if the closed-loop voltage gain is increased by steadily increasing the feedback resistor value toward infinity, what happens to the overall bandwidth and behavior of the system?

Options :

1. The bandwidth remains entirely unchanged, and the circuit begins to oscillate on its own.
2. The bandwidth expands exponentially, and the input impedance drops down to zero ohms.
3. The bandwidth shrinks toward zero, and the circuit acts as a simple open-loop voltage comparator.
4. The bandwidth expands linearly, and the circuit stays completely stable at high frequencies.

Question Number : 3 Question Id : 4410092092427 Question Type : MCQ Option Shuffling : No

Which set of parameters correctly describes the operating characteristics of a BJT common collector amplifier?

Options :

1. Low input impedance, high output impedance, and a voltage gain much greater than unity
2. High input impedance, high output impedance, and a phase shift of 180 degrees between input and output
3. Low input impedance, low output impedance, and a high negative current gain
4. High input impedance, low output impedance, and a voltage gain approximately equal to unity

Question Number : 4 Question Id : 4410092092436 Question Type : MCQ Option Shuffling : No

When two single-stage common-emitter (CE) amplifiers, each with a voltage gain of 100, are directly coupled in cascade, what will the overall voltage gain be in decibels (dB) and how will the overall bandwidth compare to that of a single stage?

Options :

1. Overall gain is 40 dB and overall bandwidth is less than single stage
2. Overall gain is 20 dB and overall bandwidth is unchanged
3. Overall gain is 80 dB and overall bandwidth is less than single stage
4. Overall gain is 20 dB and overall bandwidth is greater than single stage

Question Number : 5 Question Id : 4410092092448 Question Type : MCQ Option Shuffling : No

Which combination of operational amplifier (op-amp) circuits is sequentially connected together to simultaneously design a square wave and a triangular wave function generator system?

Options :

1. A Schmitt trigger (comparator) stage followed by an integrator stage
2. A differentiator stage followed by an inverting amplifier stage
3. A summing amplifier stage followed by a non-inverting amplifier stage
4. A logarithmic amplifier stage followed by a precision peak detector stage

Question Number : 6 Question Id : 4410092092460 Question Type : MCQ Option Shuffling : No

Identify the correct option to fill in the blanks.

In power amplifier circuits, the _____ faces the problem of crossover distortion and it can be minimized using _____.

Options :

1. Class A amplifier; a large emitter bypass capacitor
2. Class D amplifier; an external pulse-width modulation controller
3. Class C amplifier; a high-Q resonant tuned LC tank circuit
4. Class B amplifier; Class AB biasing with a small forward bias

Question Number : 7 Question Id : 4410092092466 Question Type : MCQ Option Shuffling : No

If the transient peak reverse voltage applied across a diode in a bridge rectifier circuit exceeds its rated Peak Inverse Voltage (PIV), what is the immediate electrical consequence regarding the diode's structural behaviour?

Options :

1. The diode automatically shifts into a forward-biased state to harmlessly dissipate the excess voltage.
2. The rectifier diode enters into an un-stabilized avalanche breakdown region.
3. The diode acts as a perfect open circuit to completely isolate the load resistor from the transformer.
4. The diode increases its internal depletion region width to naturally block the excess reverse current.

Question Number : 8 Question Id : 4410092092429 Question Type : MCQ Option Shuffling : No

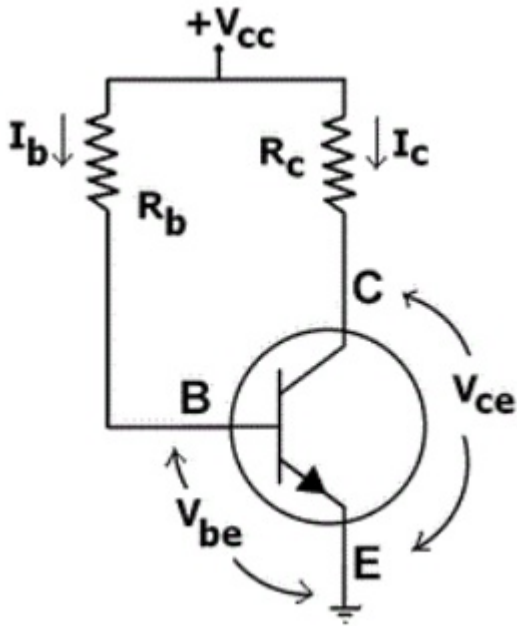
Which of the following statements correctly describes the relationship between the AC load line and the DC load line on the output characteristics of a BJT CE amplifier?

Options :

1. The AC load line is identical to the DC load line regardless of the presence of coupling capacitors, bypass capacitors, or external loads.
2. The AC load line is always perfectly horizontal because the AC signal voltage remains constant across the collector resistor.
3. The AC load line has a steeper slope than the DC load line because the AC load resistance is lower than the DC load resistance.
4. The AC load line has a gentler slope than the DC load line because the AC load resistance is higher than the DC load resistance.

Question Number : 9 Question Id : 4410092092431 Question Type : MCQ Option Shuffling : No

For a simple fixed-bias BJT common emitter circuit with a 10 V supply and a 2 k Ω collector resistor, what is the DC operating quiescent point (Q-point) when it is set exactly at the center of the DC load line?

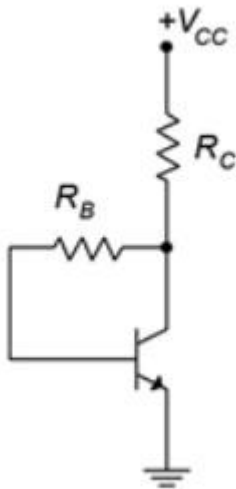


Options :

1. $V_{ce} = 5 \text{ V}$, $I_c = 5 \text{ mA}$
2. $V_{ce} = 2 \text{ V}$, $I_c = 4 \text{ mA}$
3. $V_{ce} = 7 \text{ V}$, $I_c = 1.5 \text{ mA}$
4. $V_{ce} = 5 \text{ V}$, $I_c = 2.5 \text{ mA}$

Question Number : 10 Question Id : 4410092092432 Question Type : MCQ Option Shuffling : No

In a collector feedback bias circuit, if the supply voltage is 12 V, the collector current is 2 mA, and the collector-emitter voltage is set to 60% of the supply voltage, what is the value of the collector resistor, assuming negligible base current?



Options :

1. 4.8 k Ω
2. 2.4 k Ω
3. 3.0 k Ω
4. 1.2 k Ω

Question Number : 11 Question Id : 4410092092434 Question Type : MCQ Option Shuffling : No

For a voltage-divider bias BJT network in a common emitter configuration, the emitter resistor provides _____ feedback, and to ensure the voltage gain is maximized for a small-signal amplifier, this emitter resistor should be _____.

Options :

1. DC positive, bypassed
2. DC positive, unbypassed
3. DC negative, bypassed
4. DC negative, unbypassed

Question Number : 12 Question Id : 4410092092438 Question Type : MCQ Option Shuffling : No

For a small signal single-stage CE amplifier, if the power observed at the 3dB cutoff is x watt and the approximate voltage gain at the higher cutoff point is 70.70, what are the power in the midband and the voltage gain in the midband, respectively?

Options :

1. 2x watt, 70.70
2. x watt, 50
3. x watt, 100
4. 2x watt, 100

Question Number : 13 Question Id : 4410092092446 Question Type : MCQ Option Shuffling : No

If a sinusoidal signal of 1 V peak-to-peak at 1 kHz is applied to the non-inverting terminal of an open-loop op-amp, with the inverting terminal grounded, what function does the circuit perform and what is the nature of its output?

Options :

1. Zero-crossing detector, a square wave 180 degrees out-of-phase with the input signal
2. Peak detector, a constant positive direct current voltage level
3. Integrator, a cosine wave shifted by 90 degrees
4. Zero-crossing detector, a square wave in-phase with the input signal

Question Number : 14 Question Id : 4410092092463 Question Type : MCQ Option Shuffling : No

If the RMS load current for a full wave rectifier connected to a 2 ohm load resistor is 5 A and the average DC current is 4 A, what are the ripple factor and the DC power delivered to the load?

Options :

1. 0.6, 16 W
2. 1.0, 20 W
3. 0.5, 40 W
4. 0.75, 32 W

Question Number : 15 Question Id : 4410092092518 Question Type : MCQ Option Shuffling : No

According to electromagnetic theory, what is the primary physical barrier to transmitting unmodulated audio signals (20 Hz – 20 kHz) wirelessly, due to the direct relationship between wavelength and radiating element length?

Options :

1. High signal bandwidth
2. Poor signal quality
3. Very large antenna size requirement
4. Excessive power consumption

Question Number : 16 Question Id : 4410092092523 Question Type : MCQ Option Shuffling : No

Which type of optical fiber supports only one propagation mode?

Options :

1. Multimode Graded-Index Fiber
2. Multimode Plastic Optical Fiber
3. Single-Mode Fiber
4. Multimode Step-Index Fiber

Question Number : 17 Question Id : 4410092092499 Question Type : MCQ Option Shuffling : No

In V.32 trellis-coded modulation, what is the main function of the additional redundant bit introduced in the system?

Options :

1. Synchronization of carrier frequency
2. Error detection and improved reliability
3. Increasing baud rate
4. Reducing transmission bandwidth

Question Number : 18 Question Id : 4410092092504 Question Type : MCQ Option Shuffling : No

A digital modulation scheme transmits data at 200 kbps using symbols that each carry 5 bits. What is the required symbol rate for this system?

Options :

1. 40 kbaud
2. 20 kbaud
3. 30 kbaud
4. 50 kbaud

Question Number : 19 Question Id : 4410092092509 Question Type : MCQ Option Shuffling : No

Which of the following combinations best represents the fundamental building blocks of an IoT system?

Options :

1. Servers, Switches, Repeaters, and Multiplexers
2. Sensors, Connectivity, Data Processing, and Applications and actuator control
3. Antennas, Mixers, Filters, and Detectors
4. Routers, Modems, Amplifiers, and Oscillators

Question Number : 20 Question Id : 4410092092506 Question Type : MCQ Option Shuffling : No

Which of the following statements is **NOT** correct regarding the spectrum of frequency modulation with modulation index represented by β ?

Options :

1. Increasing the amplitude of the modulating signal (with fm constant) increases both frequency deviation and FM bandwidth.
2. The number of significant sidebands generally increases as the modulation index β increases.
3. For $\beta = 4.2$, the carrier component is almost zero.
4. The total transmitted power remains constant irrespective of β .

Question Number : 21 Question Id : 4410092092511 Question Type : MCQ Option Shuffling : No

A cellular network uses a cluster size of $N=12$. The approximate co-channel reuse ratio is:

Options :

1. 3
2. 4.2
3. 6
4. 12

Question Number : 22 Question Id : 4410092092513 Question Type : MCQ Option Shuffling : No

In a cellular system, the ping-pong effect during handover is primarily caused by:

Options :

1. Rapid fluctuations in received signal strength due to multipath fading
2. Insufficient hysteresis margin and short time-to-trigger (TTT) settings
3. Rapid signal fluctuations due to multipath fading combined with inadequate hysteresis margin and short TTT settings
4. Excessively large frequency reuse distance between neighboring cells

Question Number : 23 Question Id : 4410092092519 Question Type : MCQ Option Shuffling : No

When 3×10^{11} photons each with a wavelength of 0.62 micro meter are incident on a photodiode, on average 1.2×10^{11} electrons are collected at the terminals of the device. The Quantum efficiency of the photodiode is _____

Options :

1. 12%
2. 40%
3. 0.32%
4. 80%

Question Number : 24 Question Id : 4410092092522 Question Type : MCQ Option Shuffling : No

Which planar integrated technology demultiplexes 40 or more channels simultaneously in a single parallel step without cascading discrete serial filtering components, keeping insertion loss uniform across all channels?

Options :

1. Thin-Film Filters (TFF)
2. Arrayed Waveguide Gratings (AWG)
3. Fiber Bragg Gratings (FBG)
4. Prism-based couplers

Question Number : 25 Question Id : 4410092092538 Question Type : MCQ Option Shuffling : No

Which Boolean theorem demonstrates the complement law?

Options :

1. $A + \bar{A} = 1$
2. $A . A = A$
3. $A . 1 = A$
4. $A + 0 = A$

Question Number : 26 Question Id : 4410092092561 Question Type : MCQ Option Shuffling : No

Which statement best describes the truth table of an EX-OR gate with two inputs?

Options :

1. The output is high when both inputs are high
2. The output is high when both inputs are low
3. The output is high when exactly one input is high
4. The output is always high

Question Number : 27 Question Id : 4410092092570 Question Type : MCQ Option Shuffling : No

What does overflow mean in the context of binary addition for signed numbers?

Options :

1. A carry occurs from the least significant bit
2. The result is outside the representable range for the given number of bits
3. The binary sum equals zero
4. A carry is generated at any bit position

Question Number : 28 Question Id : 4410092092542 Question Type : MCQ Option Shuffling : No

A full adder can be constructed using which combination of half adders and gates?

Options :

1. Three half adders and one EX-OR gate
2. One half adder and three OR gates
3. Two half adders and one OR gate
4. Two half adders and two EX-OR gates

Question Number : 29 Question Id : 4410092092566 Question Type : MCQ Option Shuffling : No

If the binary number 101011.101 is to be converted to decimal, what is the resulting value?

Options :

1. 43.625
2. 45.25
3. 42.375
4. 44.5

Question Number : 30 Question Id : 4410092092571 Question Type : MCQ Option Shuffling : No

In a Complex Programmable Logic Device (CPLD), consider the role of the global interconnect matrix (GIM) in relation to macrocells, product-term sharing, and device routing architecture. Which statement correctly describes its primary function?

Options :

1. It implements programmable AND-OR logic by directly combining product terms from multiple macrocells to reduce logic depth.
2. It provides a centralized routing structure that connects macrocells and function blocks while preserving deterministic timing across the device.
3. It generates global clock distribution and synchronizes all macrocells using a dedicated low-skew clock tree embedded in the interconnect matrix.
4. It stores configuration bits for programmable logic elements and dynamically reconfigures logic functions during runtime operation.

Question Number : 31 Question Id : 4410092092541 Question Type : MCQ Option Shuffling : No

In a digital circuit, if a half adder is implemented using only NAND gates, what is the minimum number of NAND gates required to realize both the sum and carry outputs?

Options :

1. Seven NAND gates
2. Six NAND gates
3. Four NAND gates
4. Five NAND gates

Question Number : 32 Question Id : 4410092092548 Question Type : MCQ Option Shuffling : No

An ideal 12-bit uniform quantizer digitizes a full-scale sinusoidal signal. Assuming quantization noise is uniformly distributed and uncorrelated with the input signal, what is the theoretical signal-to-quantization-noise ratio (SQNR) in dB?

Options :

1. 69 dB
2. 74 dB
3. 72 dB
4. 77 dB

Question Number : 33 Question Id : 4410092092551 Question Type : MCQ Option Shuffling : No

A sequential circuit is specified by the following state diagram: It has four states (S0, S1, S2, S3) and one input X. The transitions are as follows: From S0, on X=0 go to S1, on X=1 go to S2; from S1, on X=0 go to S0, on X=1 go to S3; from S2, on X=0 go to S3, on X=1 go to S0; from S3, on X=0 go to S2, on X=1 go to S1. Which one of the following statements is TRUE regarding this state diagram?

Options :

1. It contains at least one state with no outgoing transition
2. It is a Moore machine
3. Each state has exactly one outgoing transition for each input value (X = 0 and X = 1)
4. Every state has a self-loop for at least one input value

Question Number : 34 Question Id : 4410092092553 Question Type : MCQ Option Shuffling : No

A finite-state machine exhibits critical races because multiple state variables change simultaneously during state transitions. Which technique is most effective for eliminating such races?

Options :

1. Removing clock skew between flip-flops
2. Using arbitrary binary state assignments
3. Increasing flip-flop setup time margin
4. Employing state assignment with Gray code

Question Number : 35 Question Id : 4410092092554 Question Type : MCQ Option Shuffling : No

A designer selects a bipolar logic family for a synchronous counter because it provides TTL-compatible logic levels and significantly lower propagation delay than standard TTL by preventing transistor saturation using Schottky diodes. During testing, the designer observes increased power dissipation compared to standard TTL. Which logic family is being used, and what trade-off is being observed?

Options :

1. ECL family; fan-out is prioritized over voltage swing
2. DTL family; speed is sacrificed for lower power
3. RTL family; power is lowered but fan-out is limited
4. TTL Schottky family; speed is gained at the expense of higher power consumption

Question Number : 36 Question Id : 4410092092555 Question Type : MCQ Option Shuffling : No

A CMOS inverter is required to drive four identical CMOS gates. If the fan-out is increased to eight gates, which of the following will most likely occur, assuming all gates have equal input capacitance?

Options :

1. The noise margin will decrease significantly
2. The logic threshold voltage will decrease
3. The propagation delay of the inverter will increase
4. The power dissipation will decrease

Question Number : 37 Question Id : 4410092092563 Question Type : MCQ Option Shuffling : No

Which of the following statements best describes a potential pitfall when using don't care conditions for logic minimization in a safety-critical digital system?

Options :

1. Ignoring don't cares always leads to optimal minimization
2. Using don't cares has no effect on circuit reliability
3. Improperly assigning outputs for don't care conditions can lead to unintended circuit responses if those states ever occur
4. Don't care conditions always guarantee hazard-free outputs

Question Number : 38 Question Id : 4410092092578 Question Type : MCQ Option Shuffling : No

A 5-bit asynchronous ripple up-counter counts in natural binary sequence from 00000_2 (0_{10}) to 11111_2 (31_{10}). All flip-flops are ideal and assumed to settle before decoding logic evaluates outputs (no transient glitches considered). The outputs Q4 and Q2 (MSB = Q4, LSB = Q0) are connected to a 2-input AND gate. The AND gate output is connected to an asynchronous active-high reset of all flip-flops. When the counter reaches a state where the reset condition becomes true, it is immediately reset to 00000_2 . What is the resulting modulus of the counter?

Options :

1. 16
2. 20
3. 10
4. 32

Question Number : 39 Question Id : 4410092092672 Question Type : MCQ Option Shuffling : No

A $50\ \Omega$ transmission line is connected directly to a resonant half-wave dipole antenna. What is the most likely effect?

Options :

1. Increase in antenna gain due to improved radiation efficiency
2. Impedance mismatch resulting in reflection loss
3. Change in resonant frequency due to altered current distribution
4. Increase in radiation resistance due to direct line connection

Question Number : 40 Question Id : 4410092092691 Question Type : MCQ Option Shuffling : No

For a uniform plane wave travelling in free space, the electric field intensity is $E = 120\pi$ V/m. The corresponding magnetic field intensity is:

Options :

1. 1 A/m
2. 120π A/m
3. π A/m
4. $1/(120\pi)$ A/m

Question Number : 41 Question Id : 4410092092680 Question Type : MCQ Option Shuffling : No

In a cavity magnetron, let V_{0c} denote the Hull cutoff voltage and B_{0c} denote the Hull cutoff magnetic field. Assume V_0 is the operating anode voltage, and B_0 is the operating magnetic field, Under which condition will electrons fail to reach the anode?

Options :

1. $V_0 > V_{0c}$ for a given B_0 , and $B_0 < B_{0c}$ for a given V_0
2. $V_0 = V_{0c}$ for a given B_0 , and $B_0 = B_{0c}$ for a given V_0
3. $V_0 < V_{0c}$ for a given B_0 , and $B_0 > B_{0c}$ for a given V_0
4. $V_0 > V_{0c}$ for a given B_0 , and $B_0 > B_{0c}$ for a given V_0

Question Number : 42 Question Id : 4410092092682 Question Type : MCQ Option Shuffling : No

A uniform plane wave propagating in free space has an electric field $\mathbf{E} = 48e^{-j\beta z} \mathbf{a}_y$ V/m. The time-average Poynting vector is:

Options :

1. $(9.6/\pi) \mathbf{a}_z$ W/m²
2. $-(9.6/\pi) \mathbf{a}_z$ W/m²
3. $(4.8/\pi) \mathbf{a}_z$ W/m²
4. $-(4.8/\pi) \mathbf{a}_z$ W/m²

Question Number : 43 Question Id : 4410092092687 Question Type : MCQ Option Shuffling : No

A lossless transmission line has a characteristic impedance of 50Ω . The standing-wave ratio on the line is 4, and the load end corresponds to a voltage minimum. Which combination of load resistance (R_L) and reflection coefficient (Γ) is correct?

Options :

1. $R_L = 12.5 \Omega$, $\Gamma = -0.6$
2. $R_L = 200 \Omega$, $\Gamma = +0.6$
3. $R_L = 0 \Omega$, $\Gamma = -1$
4. $R_L = 50 \Omega$, $\Gamma = 0$

Question Number : 44 Question Id : 4410092092689 Question Type : MCQ Option Shuffling : No

A spherically symmetric vector field is represented by $\mathbf{F} = K r^n \mathbf{a}_r$, where K is a constant and $r = \sqrt{(x^2 + y^2 + z^2)}$. If the field is solenoidal, the value of n is:

Options :

1. 0
2. -1
3. 1
4. -2

Question Number : 45 Question Id : 4410092092692 Question Type : MCQ Option Shuffling : No

A plane electromagnetic wave is normally incident from free space onto a lossless medium having $\epsilon = 4\epsilon_0$ and $\mu = 9\mu_0$. Which combination correctly describes the transmitted wave?

Options :

1. Velocity becomes $c/6$, wavelength becomes $\lambda/6$, frequency remains unchanged.
2. Velocity becomes $c/6$, wavelength remains unchanged, frequency becomes $f/6$.
3. Velocity becomes $c/3$, wavelength becomes $\lambda/3$, frequency remains unchanged.
4. Velocity is unchanged, wavelength is unchanged, frequency is halved

Question Number : 46 Question Id : 4410092092653 Question Type : MCQ Option Shuffling : No

Which temperature sensor is represented by its negative temperature coefficient (NTC) characteristics?

Options :

1. Type-J Thermocouple
2. Platinum RTD
3. Thermistor
4. Radiation pyrometer

Question Number : 47 Question Id : 4410092092658 Question Type : MCQ Option Shuffling : No

In a porous silicon humidity sensor, what is mainly responsible for its high sensitivity to humidity?

Options :

1. Low dielectric constant of silicon
2. High thermal conductivity of silicon
3. High surface-to-volume ratio of porous silicon
4. High melting point of silicon

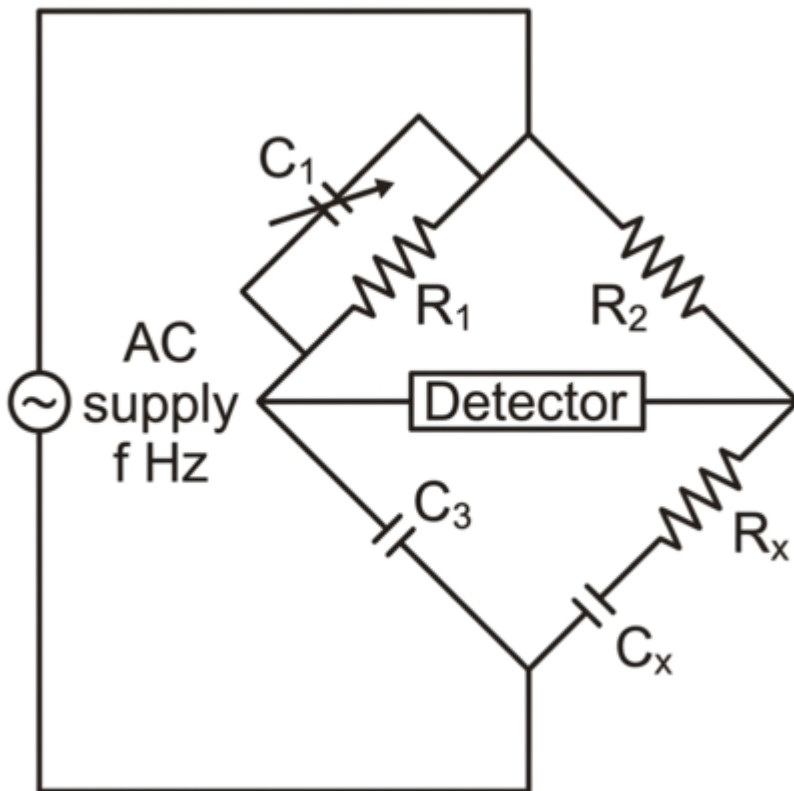
Question Number : 48 Question Id : 4410092092646 Question Type : MCQ Option Shuffling : No

Which bridge is particularly suitable for the precise measurement of low self-inductance and low Q-factor coils?

Options :

1. Anderson Bridge
2. Kelvin Bridge
3. Schering Bridge
4. De-Sauty Bridge

Question Number : 49 Question Id : 4410092092647 Question Type : MCQ Option Shuffling : No
In a Schering Bridge (as shown in figure), if the known capacitor C_3 is 1000 pF, the known resistor R_2 is 10 k Ω , and the balancing resistor R_1 is 500 k Ω , what is the value of the unknown capacitor C_x ?



Options :

1. 50 nF
2. 75 nF
3. 25 nF
4. 100 nF

Question Number : 50 Question Id : 4410092092650 Question Type : MCQ Option Shuffling : No
Which MEMS application is most suitable for detecting static tilt relative to gravity to trigger screen auto-rotation in mobile devices?

Options :

1. MEMS microphones in phones
2. Accelerometers in smartphones
3. Pressure sensors in devices
4. Gyroscopes in tablets

Question Number : 51 Question Id : 4410092092651 Question Type : MCQ Option Shuffling : No

A CRT has an anode voltage of 2500 V and parallel deflecting plates 2 cm long and 4 mm apart. The screen is 40 cm from the centre of the plates. The beam is required to be deflected by 4 cm on the screen. The input voltage is applied through amplifiers having an overall gain of 50. The required input voltage is:

Options :

1. 1 V
2. 2 V
3. 4 V
4. 3 V

Question Number : 52 Question Id : 4410092092661 Question Type : MCQ Option Shuffling : No

Which statements are correct regarding the dynamic behavior of a capacitive transducer used for dynamic displacement measurement?

1. High-frequency displacement variations are transmitted with less attenuation.
2. Low-frequency components may suffer attenuation.
3. The transducer behaves similarly to a low-pass filter.

Options :

1. 1, 2 and 3
2. 1 and 3 only
3. 2 and 3 only
4. 1 and 2 only

Question Number : 53 Question Id : 4410092092663 Question Type : MCQ Option Shuffling : No

A differential amplifier has a common-mode input voltage of 500 mV and a differential-mode input voltage of 30 mV. The corresponding output voltages are 5 mV and 3 V, respectively.

Consider the following statements:

1. The differential-mode gain of the amplifier is 100.
2. The common-mode gain of the amplifier is 0.01.
3. The common-mode rejection ratio (CMRR) of the amplifier is 10,000.

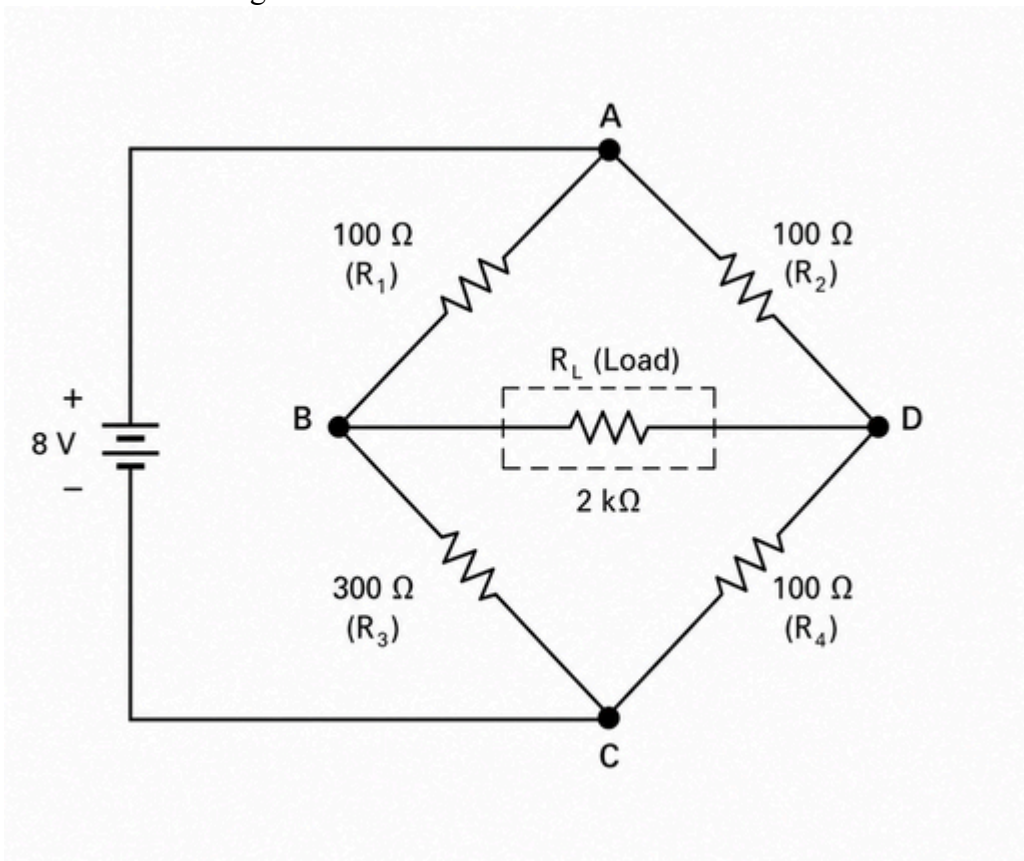
Which of the above statements are correct?

Options :

1. 1 and 2 only
2. 2 and 3 only
3. 1 and 3 only
4. 1, 2 and 3

Question Number : 54 Question Id : 4410092092665 Question Type : MCQ Option Shuffling : No

In a Wheatstone bridge as shown in figure, three resistors are $100\ \Omega$ each and the fourth resistor is $300\ \Omega$. The bridge is supplied with 8 V . If the output terminals are connected to a $2\text{ k}\Omega$ resistor, how much current flows through this resistor?



Options :

1. $1000\ \mu\text{A}$
2. $250\ \mu\text{A}$
3. $100\ \mu\text{A}$
4. $500\ \mu\text{A}$

Question Number : 55 Question Id : 4410092092702 Question Type : MCQ Option Shuffling : No

In the static Voltage Transfer Characteristic (VTC) of an ideal CMOS inverter, what is the output voltage (V_{out}) when the input voltage (V_{in}) equals the supply voltage V_{DD} ?

Options :

1. $V_{\text{DD}} / 2$
2. V_{DD}
3. 0 V
4. $V_{\text{Th},n}$

Question Number : 56 Question Id : 4410092092707 Question Type : MCQ Option Shuffling : No

Which thin film deposition method is characterized by alternating, self-limiting surface reactions that deposit precisely one atomic layer per cycle?

Options :

1. Atomic layer deposition (ALD)
2. Low-pressure chemical vapor deposition (LPCVD)
3. Molecular beam epitaxy (MBE)
4. Sputter deposition

Question Number : 57 Question Id : 4410092092696 Question Type : MCQ Option Shuffling : No

During the silicon chemical vapor deposition (CVD) epitaxial growth process using silicon tetrachloride (SiCl_4) as the source gas, hydrogen chloride (HCl) gas is actively generated as a reaction byproduct or intentionally introduced. What is the deterministic role of this (HCl) component within the chemical equilibrium of the epitaxial reactor?

Options :

1. It acts as a reducing agent that actively prevents the oxidation of the silicon surface by residual oxygen or water vapor.
2. It serves as an interstitial n-type dopant source that automatically establishes the base resistivity profile of the growing epitaxial layer.
3. It functions as a catalytic cracking agent that lowers the thermal activation energy required to dissociate the native silicon dioxide layer.
4. It establishes a reversible chemical etching reaction that selectively removes misaligned or thermodynamically unstable silicon atoms from defect sites.

Question Number : 58 Question Id : 4410092092701 Question Type : MCQ Option Shuffling : No

Under static operating conditions with the input voltage held steady at the positive supply rail ($V_{in} = V_{DD}$), a conventional CMOS inverter ideally exhibits no direct current path between the supply rails. Which transistor operating condition is responsible for this behavior?

Options :

1. The gate oxide capacitances of the NMOS and PMOS devices prevent DC conduction.
2. The PMOS transistor is in cut-off, leaving the pull-up path non-conducting.
3. The NMOS transistor operates in the linear region, increasing resistance to ground.
4. The PMOS transistor enters velocity saturation, limiting channel current.

Question Number : 59 Question Id : 4410092092694 Question Type : MCQ Option Shuffling : No

A surface-channel MOS capacitor element within a three-phase, n-channel Charge-Coupled Device (CCD) is fabricated on a uniformly doped p-type silicon substrate. The gate oxide (SiO_2) layer has a capacitance per unit area of $C_{ox} = 2.0 \times 10^{-7} \text{ F/cm}^2$. The substrate doping and material parameters yield a depletion coefficient factor of:

$$\sqrt{2q\epsilon_s N_a} = 2.0 \times 10^{-7} \text{ C/(V}^{1/2} \cdot \text{cm}^2\text{)}.$$

The flat-band voltage V_{FB} of this MOS configuration is exactly -1.0 V . Calculate the surface potential Φ_s immediately after a clock pulse of $V_G = +5.0 \text{ V}$ is applied to the gate, assuming the structure is in a state of complete deep depletion where no minority carrier inversion charge has accumulated.

Options :

1. 2.0 V
2. 4.0 V
3. 5.0 V
4. 6.0 V

Question Number : 60 Question Id : 4410092092706 Question Type : MCQ Option Shuffling : No

An NMOS transistor features a baseline threshold voltage $V_{th0} = 0.40 \text{ V}$ under a zero body-bias condition ($V_{BS} = 0 \text{ V}$). The internal physical characteristics of the device include a bulk substrate surface potential component of $|2\Phi_F| = 0.64 \text{ V}$ and a body-effect parameter of $\gamma = 0.50 \text{ V}^{1/2}$. If a reverse body-bias potential is applied across the substrate terminal such that the source-to-substrate voltage becomes $V_{SB} = 1.60 \text{ V}$, the value of the newly adjusted threshold voltage (V_{th}) is closest to which of the following? Assume square root of $2.24 = 1.5$

Options :

1. 1.15 V
2. 0.55 V
3. 0.40 V
4. 0.75 V

Question Number : 61 Question Id : 4410092092714 Question Type : MCQ Option Shuffling : No

An NMOS transistor in a 65 nm CMOS process has a gate oxide capacitance per unit area of $C_{ox} = 5.0 \text{ fF}/\mu\text{m}^2$, a gate width of $W = 0.4 \mu\text{m}$, and a gate-to-drain overlap length of $L_{ov} = 0.05 \mu\text{m}$. The drain diffusion region has an area junction capacitance density of $2.0 \text{ fF}/\mu\text{m}^2$ over a drain area of $0.1 \mu\text{m}^2$, and a sidewall junction capacitance density of $0.5 \text{ fF}/\mu\text{m}$ over a sidewall perimeter of $0.6 \mu\text{m}$ (excluding the gate edge). Neglecting gate-to-channel capacitance, what is the total drain-associated parasitic capacitance, summing the gate-to-drain overlap capacitance, the drain area junction capacitance, and the drain sidewall junction capacitance?

Options :

1. 0.6 fF
2. 0.3 fF
3. 0.4 fF
4. 0.5 fF

Question Number : 62 Question Id : 4410092172573 Question Type : MCQ Option Shuffling : No

In an 8051 microcontroller, the Accumulator (A) contains decimal value 12, and register R1 contains decimal value 23. If the instruction ADD A, R1 is executed, what will be the contents of the Accumulator after execution (assuming no overflow occurs)?

Options :

1. A contains 255
2. A contains 35
3. R1 contains 255
4. R1 contains 35

Question Number : 63 Question Id : 4410092172596 Question Type : MCQ Option Shuffling : No

A student needs to rearrange the bits in the Accumulator so that the high nibble becomes the low nibble and vice versa, but wants the status flags to remain unchanged. Which 8051 instruction should the student use?

Options :

1. XCHD A, @Ri
2. SWAP A
3. ANL A, #0Fh
4. ADD A, #0Fh

Question Number : 64 Question Id : 4410092172580 Question Type : MCQ Option Shuffling : No

A sensor-based microcontroller system monitors both pressure and humidity. The system's interrupt priority controller must decide which signal to attend when both interrupts arrive simultaneously, given that pressure changes can destabilize the entire system more rapidly than humidity fluctuations. What is the most appropriate rationale for assigning higher interrupt priority in this scenario?

Options :

1. Assigning priority based on sensor installation date provides scheduling fairness
2. Prioritizing pressure ensures system stability during critical fluctuations
3. Equal priority for both interrupts prevents any signal from being overlooked
4. Prioritizing humidity reflects its environmental significance for long-term trends

Question Number : 65 Question Id : 4410092172584 Question Type : MCQ Option Shuffling : No

A sensor-based irrigation unit is designed to independently adjust water flow based on soil moisture and ambient temperature, without any user intervention. Which attribute most fundamentally distinguishes this unit as an embedded system rather than a general-purpose computing system?

Options :

1. It performs a pre-defined, application-specific function autonomously
2. It provides a user interface for multiple tasks
3. It processes external commands in real time for diverse applications
4. It can be reprogrammed frequently for new, unrelated roles

Question Number : 66 Question Id : 4410092172590 Question Type : MCQ Option Shuffling : No

Which component is typically required to correctly interface an RS-232 signal with TTL logic circuits?

Options :

1. An optocoupler without supporting circuitry
2. A level shifter or RS-232 to TTL converter
3. A direct wire connection
4. A resistor voltage divider only

Question Number : 67 Question Id : 4410092172576 Question Type : MCQ Option Shuffling : No

Which of the following best describes a key risk when incomplete context preservation occurs during nested interrupts in assembly programming?

Options :

1. Critical register values may be lost, leading to unpredictable behavior after return.
2. Stack overflow will always occur, causing system reset.
3. The code size increases significantly, affecting performance.
4. Interrupts will execute faster due to less state saving.

Question Number : 68 Question Id : 4410092172577 Question Type : MCQ Option Shuffling : No

In an embedded timer-based periodic scheduler, which method best minimizes timing jitter for task release?

Options :

1. Checking a status register after main code execution
2. Using a hardware timer interrupt to trigger the task release at each precise timer overflow
3. Resetting the timer manually in the main loop
4. Polling timer overflow flag in the main program loop

Question Number : 69 Question Id : 4410092172585 Question Type : MCQ Option Shuffling : No

Which system architecture is most suitable for a large-scale environmental monitoring system with hundreds of embedded sensor nodes requiring scalable communication, local data aggregation, fault tolerance, and real-time event processing?

Options :

1. Hierarchical clustered architecture with distributed gateways
2. Single central gateway with direct sensor node connections
3. Peer-to-peer mesh with all nodes communicating equally
4. Flat architecture with all nodes communicating to a central server

Question Number : 70 Question Id : 4410092172592 Question Type : MCQ Option Shuffling : No

A team is designing an automated telescope mount using a microcontroller to interface with a hybrid stepper motor. The controller must achieve precise angular positioning for astronomical tracking. Mechanical testing confirms that the motor provides adequate holding torque, the stepping sequence has been verified as correct, and no electromagnetic interference or system resets are observed. However, after several hours of continuous operation, cumulative azimuth errors still occur. Which interfacing issue most plausibly explains this behavior?

Options :

1. The software-based step pulse generation introduces irregular pulse timing due to non-deterministic interrupt latency, causing missed or extra steps during long observation sessions.
2. The motor driver uses insufficient holding current during idle periods, leading to gradual mechanical slippage under telescope load.
3. The microcontroller implements the incorrect stepping sequence, resulting in the motor drifting from its intended path over time.
4. The stepper motor interface employs unshielded cabling, causing sporadic electromagnetic interference that occasionally resets the system.

Question Number : 71 Question Id : 4410092092611 Question Type : MCQ Option Shuffling : No

In Norton's theorem, if the Norton resistance is 2Ω and the load resistance is 4Ω , what fraction of the Norton current flows through the load?

Options :

1. $\frac{1}{2}$ of the Norton current
2. $\frac{1}{3}$ of the Norton current
3. $\frac{1}{6}$ of the Norton current
4. $\frac{1}{4}$ the Norton current

Question Number : 72 Question Id : 4410092092636 Question Type : MCQ Option Shuffling : No

A system function has poles at $z = 0.5$ and $z = 0.9$. For a causal and stable system, what is the ROC?

Options :

1. ROC is $|z|$ less than 0.5
2. ROC is 0.5 less than $|z|$ less than 0.9
3. ROC is $|z|$ greater than 0.9
4. ROC is $|z|$ less than 0.9

Question Number : 73 Question Id : 4410092092614 Question Type : MCQ Option Shuffling : No

Three branches are connected in parallel between terminals **A** and **B**. Each branch consists of an ideal voltage source with its **positive terminal connected to A** and a series resistance. The branch parameters are:

- Branch 1: 12 V in series with $2\ \Omega$
- Branch 2: 6 V in series with $3\ \Omega$
- Branch 3: 18 V in series with $6\ \Omega$

Using **Millman's theorem**, what is the equivalent source voltage across terminals **A** and **B**?

Options :

1. 10 V
2. 11 V
3. 6.5 V
4. 7.5 V

Question Number : 74 Question Id : 4410092092615 Question Type : MCQ Option Shuffling : No

A signal contains frequency components from 0 to 5 kHz. What is the **maximum sampling interval (Nyquist sampling interval)** that avoids aliasing?

Options :

1. 0.25 ms
2. 0.2 ms
3. 0.1 ms
4. 0.15 ms

Question Number : 75 Question Id : 4410092092621 Question Type : MCQ Option Shuffling : No

If $x(t) = \sin(t) + t$, what is the odd part of $x(t)$?

Options :

1. $x_{\text{odd}}(t) = t$
2. $x_{\text{odd}}(t) = \sin(t) + t$
3. $x_{\text{odd}}(t) = 0$
4. $x_{\text{odd}}(t) = \sin(t)$

Question Number : 76 Question Id : 4410092092623 Question Type : MCQ Option Shuffling : No

For the system $dx/dt = Ax + Bu$, $y = Cx + Du$, if A is a 3×3 matrix and B is 3×1 , what must be the dimensions of C for a single-output system?

Options :

1. 3×3
2. 1×1
3. 1×3
4. 3×1

Question Number : 77 Question Id : 4410092092608 Question Type : MCQ Option Shuffling : No
A planar circuit has **four meshes**. An ideal current source is common to **Mesh 2 and Mesh 3**, forming a **supermesh**. There are no other current sources in the circuit. Using mesh analysis, how many **KVL equations** and **constraint equations** are required to determine all mesh currents?

Options :

1. 4 KVL equations and 0 constraint equations
2. 2 KVL equations and 2 constraint equations
3. 4 KVL equations and 1 constraint equation
4. 3 KVL equations and 1 constraint equation

Question Number : 78 Question Id : 4410092092619 Question Type : MCQ Option Shuffling : No
For a causal IIR system to be stable, what must be true regarding the positions of all its poles in the z-plane?

Options :

1. All poles must be strictly inside the unit circle for the system to be stable.
2. At least one pole must be on the unit circle for bounded input bounded output stability.
3. Stability is guaranteed as long as all zeros are inside the unit circle.
4. The system is stable if all poles are at the origin regardless of other poles.

Question Number : 79 Question Id : 4410092092632 Question Type : MCQ Option Shuffling : No
Which of the following transfer functions has the same DC gain as $H(s) = 1/(s + 2)$ but a faster transient response?

Options :

1. $H(s) = 0.5/(s + 1)$
2. $H(s) = 2.5/(s + 5)$
3. $H(s) = 1/(s + 6)$
4. $H(s) = 5/(s + 2)$

Question Number : 80 Question Id : 4410092092634 Question Type : MCQ Option Shuffling : No
For $X(s) = (s + 4)/(s^2 + 4s + 13)$, find $x(t)$ for $t \geq 0$.

Options :

1. $x(t) = e^{-2t} \sin(3t) + 4 e^{-2t} \cos(3t)$
2. $x(t) = e^{-2t} \cos(3t) - 2 e^{-2t} \sin(3t)$
3. $x(t) = \cos(3t) e^{2t} + 2 \sin(3t) e^{2t}$
4. $x(t) = e^{-2t} \cos(3t) + \frac{2}{3} e^{-2t} \sin(3t)$

Question Number : 81 Question Id : 4410092092640 Question Type : MCQ Option Shuffling : No
If a two-port network is reciprocal, which relationship among h-parameters must be satisfied?

Options :

1. $h_{12} h_{21} = h_{11} h_{22} - 1$
2. $h_{12} + h_{21} = 0$
3. $h_{12} = h_{21}$
4. $h_{11} = h_{22}$

Question Number : 82 Question Id : 4410092092641 Question Type : MCQ Option Shuffling : No

A **reciprocal** and **matched** two-port network is inserted between a matched source and a matched load. The reverse transmission coefficient is

$$S_{12} = \frac{1}{10}.$$

What is the insertion loss of the network?

Options :

1. 10 dB
2. 20 dB
3. 6 dB
4. 15 dB

Question Number : 83 Question Id : 4410092092407 Question Type : MCQ Option Shuffling : No

A boost converter operates at 1 kHz with an input voltage of 48 V and delivers an average output voltage of 80 V. Assuming ideal operation of the converter, the duty ratio of the converter is:

Options :

1. 0.2
2. 0.4
3. 0.6
4. 0.8

Question Number : 84 Question Id : 4410092092415 Question Type : MCQ Option Shuffling : No

Why would using a PLC be advantageous over traditional relay systems in a manufacturing factory, which needs to manufacture a wide variety of products?

Options :

1. PLCs operate faster than traditional relay systems, reducing production time.
2. PLCs are more compact than traditional relay systems, saving installation space.
3. PLCs are more reliable than traditional relay systems, minimizing breakdowns.
4. PLCs allow program changes without hardware modifications, enabling flexible control.

Question Number : 85 Question Id : 4410092092395 Question Type : MCQ Option Shuffling : No

In a closed-loop temperature control system for an electric furnace, which of the following is the function of the error signal generated inside the system?

Options :

1. It measures the accuracy of the temperature sensor being used for feedback in the furnace.
2. It indicates the change in load on the power supply of the system after the corrective action.
3. It monitors external disturbances affecting the furnace and changes the desired temperature accordingly.
4. It determines the corrective action to be sent to the heater to adjust the furnace temperature.

Question Number : 86 Question Id : 4410092092406 Question Type : MCQ Option Shuffling : No

A battery-operated electric vehicle uses a DC–DC converter-fed DC motor for traction. During a stop (applying the brakes), how does the traction system ensure that regenerative braking happens?

Options :

1. By keeping the supply voltage lower than the induced emf in the DC motor
2. By keeping the field voltage constant during the braking period of the DC motor
3. By increasing the supply voltage higher than the induced emf in the DC motor
4. By disconnecting the load and only using the field winding of the DC motor for operation

Question Number : 87 Question Id : 4410092092414 Question Type : MCQ Option Shuffling : No

In a Uninterruptible Power Supply (UPS) system, on which the inverter operates only during supply failure, what is the function of the rectifier under normal operating conditions?

Options :

1. To directly supply the load during normal operation
2. To maintain the full charge of the battery
3. To switch the load to the inverter continuously
4. To provide backup power during supply failure

Question Number : 88 Question Id : 4410092092396 Question Type : MCQ Option Shuffling : No

A unity-feedback system with a plant gain $G(s)$ as given below is cascaded with a proportional controller of gain K_p .

$$G(s) = \frac{5}{s(s+5)}$$

Which of the following can be the possible value of K_p such that the system is underdamped in nature?

Options :

1. 0.5
2. 1
3. 1.5
4. 0

Question Number : 89 Question Id : 4410092092409 Question Type : MCQ Option Shuffling : No

Which of the following characteristics makes the enhancement-type MOSFETs preferred for switching in power electronics applications compared to depletion-type MOSFETs?

Options :

1. They have lower on-state resistance than depletion-type MOSFETs.
2. They remain off at zero gate voltage, unlike the depletion-type MOSFETs.
3. They remain on at zero gate voltage, unlike the depletion-type MOSFETs.
4. They can use a negative gate voltage to turn on, unlike the depletion-type MOSFETs.

Question Number : 90 Question Id : 4410092092413 Question Type : MCQ Option Shuffling : No

An ideal boost converter is used to step up a DC input voltage of 24 V. The duty cycle of the switch is 0.6, and the converter operates in continuous conduction mode (CCM). If the load draws a current of 2 A, what will be the value of average inductor current for this converter?

Options :

1. 2 A
2. 3 A
3. 4 A
4. 5 A

Question Number : 91 Question Id : 4410092092419 Question Type : MCQ Option Shuffling : No

For a unity-feedback system with open-loop transfer function $G(s) = \frac{K}{s(s+2)(s+5)}$, find the range of values of K for which the closed-loop system would be stable.

Options :

1. $K > 0$
2. $K < 30$
3. $K < 0$ and $K > 50$
4. $0 < K < 70$

Question Number : 92 Question Id : 4410092092421 Question Type : MCQ Option Shuffling : No

The open-loop transfer function of a unity feedback system is given as:

$$G(s) = \frac{40}{s(1 + 0.1s)(1 + 0.4s)}$$

Which of the following is the phase crossover frequency of this system (in rad/s)?

Options :

1. 3 rad/s
2. 5 rad/s
3. 8 rad/s
4. 10 rad/s

Question Number : 93 Question Id : 4410092133736 Question Type : MCQ Option Shuffling : No

Which of the following causes the increase in electron concentration in the conduction band of a N type semiconductor?

Options :

1. Significant rise in temperature
2. Reduction of donor impurities
3. Increasing the band gap
4. Electron-hole recombination

Question Number : 94 Question Id : 4410092092593 Question Type : MCQ Option Shuffling : No

What is the function of the heterojunction at the interface between the two semiconductor materials in HEMT?

Options :

1. To block the flow of holes into the channel region
2. To create a two-dimensional electron gas channel with high mobility
3. To increase the thermal conductivity of the device
4. To prevent gate leakage current

Question Number : 95 Question Id : 4410092092598 Question Type : MCQ Option Shuffling : No

Which of the following correctly describes and explains the effect of an increase in temperature on the electrical conductivity of metals and semiconductors?

Options :

1. Conductivity remains unchanged in both due to compensation effects.
2. Both metals and semiconductors decrease conductivity as collisions rise.
3. In metals increased lattice vibrations scatter electrons while in semiconductors more electrons are excited to the conduction band.
4. Both metals and semiconductors increase conductivity due to more free electrons.

Question Number : 96 Question Id : 4410092092591 Question Type : MCQ Option Shuffling : No

Which of the following is the advantage of modulation doping in a high electron mobility transistor (HEMT) compared to uniformly doped structures when applied to high-frequency amplifiers?

Options :

1. It increases carrier recombination, enhancing gain at low frequencies.
2. It lowers the breakdown voltage, improving device reliability.
3. It increases the bandgap, allowing higher output voltage swings.
4. It reduces ionized impurity scattering, thereby enabling higher electron mobility in the channel.

Question Number : 97 Question Id : 4410092092592 Question Type : MCQ Option Shuffling : No

Which material system is **classically used** to form the two-dimensional electron gas (2DEG) in **HEMTs for low-noise microwave and satellite communication receivers**?

Options :

1. Si/Cu structure
2. AlGaAs/GaAs heterostructure
3. Si/SiGe heterostructure
4. GaN/Cu structure

Question Number : 98 Question Id : 4410092092595 Question Type : MCQ Option Shuffling : No

Which of the following conductor parameters is important to minimize signal distortion for ultra-high-frequency signal lines in telecommunications?

Options :

1. Dielectric breakdown strength of the material
2. Skin depth determined by conductivity and magnetic permeability
3. Thermal conductivity for heat dissipation
4. Work function for electron emission

Question Number : 99 Question Id : 4410092092600 Question Type : MCQ Option Shuffling : No

A green LED typically emits light with a wavelength around 530 nm. To design a blue LED, what should be bandgap required for the design of blue LED compared to that used for the green LED?

Options :

1. The blue LED bandgap should be smaller than that of the green LED
2. The blue LED bandgap should be negative compared to the green LED
3. The blue LED bandgap should be larger than that of the green LED
4. The blue LED bandgap should be identical to the green LED

Question Number : 100 Question Id : 4410092092605 Question Type : MCQ Option Shuffling : No

If a Zener diode is rated at 10 V and 1 W, what is the maximum current that can safely flow through the Zener diode without exceeding its power rating?

Options :

1. 0.1 A to avoid exceeding Zener power rating
2. 0.05 A to maintain constant voltage
3. 1.0 A for maximum load capability
4. 0.01 A for minimal heat generation